Why Ensembles Work

Diversity turns many imperfect models into one stronger model

Ensemble Methods · Foundations of Machine Learning

Diversity Is the Essential Ingredient

Identical errors

Perfectly correlated models fail on the same cases; averaging changes nothing.

Different errors

Less-correlated mistakes partially cancel when predictions are combined.

How algorithms create it

Bootstrap samples, feature subsets, and corrective reweighting manufacture diversity.

Variance of an Averaged Ensemble

Independent models — When $\rho = 0$, variance shrinks by a factor of B.

Correlated models — Positive error correlation creates a nonzero floor.

Design implication — Lowering ρ can matter more than merely adding estimators.

$$\mathrm{Var}\left(\frac{1}{B}\sum_b \hat{f}_b\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2 \longrightarrow \rho\sigma^2$$

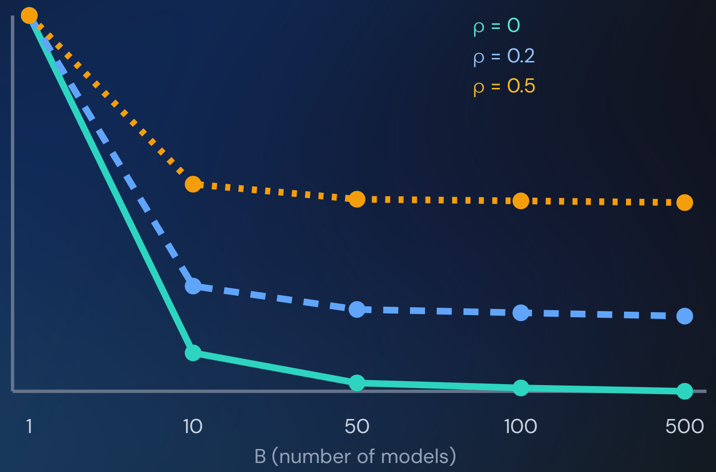
Worked example: $\sigma^2 = 4$ for a single tree, $\rho = 0.2$ correlation between trees, B = 50. Variance = 0.2·4 + (0.8·4)/50 = 0.8 + 0.064 = 0.864 — most of the reduction already happened; more trees barely help further.

Correlation Sets the Variance Floor

$\rho = 0$ — Variance keeps falling toward zero.

$\rho = 0.2$ — More models help, then flatten near $0.2\sigma^2$.

$\rho = 0.5$ — A highly correlated ensemble has little room to improve.



Majority Vote Beats Any Single Voter

Setup — B independent classifiers, each correct with probability $p > 0.5$.

Majority correct — Sum a Binomial(B, p) tail for at least $\lceil B/2 \rceil$ correct votes.

$B \to \infty$ — Majority-vote accuracy converges to 1, however small the edge over chance.

$$P(\text{majority correct}) = \sum_{k=\lceil B/2 \rceil}^{B} \binom{B}{k} p^k (1-p)^{B-k}$$

Worked example: p = 0.7, B = 3. Majority correct means 2 or 3 of 3 right: `3·0.7²·0.3 + 0.7³ = 0.441 + 0.343 = 0.784` — better than any single 70%-accurate voter. With B = 11 independent voters at p = 0.7, majority accuracy rises to roughly 0.94.

Bias–Variance, Revisited

Bagging / random forests — Start with a low-bias, high-variance deep tree and reduce variance.

Boosting — Start with high-bias, low-variance stumps and correct systematic error.

One framework — Both methods move a learner toward lower expected test error.

1. Base learner
2. Manufacture diversity
3. Combine
4. Lower test error

The Limits of Ensembling

Noise floor

No ensemble can remove irreducible uncertainty in the data.

Diminishing returns

Additional estimators eventually add little accuracy.

Latency and compute

Hundreds of models cost more to train and query than one.

Interpretability

A forest is harder to explain than a single tree.

Why Ensembles Work

Diversity

Errors must differ.

Correlation

Sets the variance floor.

Tradeoffs

Noise, cost, and opacity remain.

Next: unsupervised learning—finding structure without labels.